

CONDITIONING A MEASURE-VALUED DIFFUSION ON THE WASSERSTEIN SPACE

FIRST LAST

ABSTRACT. A concise abstract summarizing the problem, main results, and significance.

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1. INTRODUCTION

Martin: I started reading a bit about Dirichlet-Ferguson diffusion and I still don't really understand if/how it depends on the choice of metric on the space of probability densities. It is not Brownian motion for the Otto metric (as this can't be defined), but is there some dependence on the metric? I didn't understand the answers that I either got by chatGPT or found in papers. Do you have a better understanding of that? E.g. would we get a different thing if we would want to construct a more Wasserstein-Fisher-Rao type Dirichlet form? Ok, actually this is done in Lorenzo Dello Schiavo and Giacomo Enrico Sodini (2025). The Hellinger–Kantorovich metric measure geometry on spaces of measures.

In trying to further understand these Dirichlet forms constructions: Can we construct these on any infinite dimensional manifold? How would it look like in an abstract situation $\mathcal{M}, G$. Are there any particular properties of either $\mathcal{M}$ or G that we need? What about some of the standard examples, i.e.,

- (1) ℓ^2 with standard ℓ^2 -inner product as a metric (flat space)
- (2) ℓ^2 with some conformal weighted ℓ^2 -inner product (not flat anymore and no explicit geodesics)
- (3) Space of curves/surfaces with reparam invariant Sob metrics
- (4) Diffeomorphism group with Sobolev metrics

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(5) Volume preserving diffeos with L^2 -metric (i.e., Dirichlet form for the incompressible Euler equations)

Mao: My understanding is that to define a "natural" Dirichlet form (i.e., integration of the gradient), then you need a gradient on the space, and that depends on your metric. As long as you have a space with a gradient and some reference measure, you should be able to define a Dirichlet form.

2. PRELIMINARIES

2.1. Markov semigroups and generators. On $\mathbb{R}^d$, Brownian motion $\{B_t\}_{t \geq 0}$ is a prototypical time-homogeneous Markov process. If $(X_t)_{t \geq 0}$ is a (time-homogeneous) Markov process on a measurable space $(E, \mathcal{B})$, its transition probabilities induce a family of linear operators $(P_t)_{t \geq 0}$ on a suitable space of functions by

$$(2.1) \quad P_t f(x) := \int_E f(y) p_t(x, dy) := \mathbb{E}_x[f(X_t)].$$

Under standard regularity assumptions (e.g. the Feller property on a locally compact state space [4, Theorem A.2.2]), one can also construct a Markov process (or, to be exact, a Hunt process) from such a semigroup.

If $(P_t)_{t \geq 0}$ is a strongly continuous contraction semigroup on a Hilbert space, there is a one correspondence [4, Lemma 1.3.2] with a non-positive definite self-adjoint operator L called its *generator*, defined by

$$(2.2) \quad Lf := \lim_{h \downarrow 0} \frac{P_h f - f}{h}.$$

For Brownian motion on $\mathbb{R}^d$, it is the half of the Laplacian operator $L = \frac{1}{2}\Delta$. Therefore, Brownian motion can be equivalently characterized as the Markov process with generator $\frac{1}{2}\Delta$.

A natural idea for defining a "Brownian motion" on the Wasserstein space $P_2(M)$ over a Riemannian manifold M is to use the formal Riemannian structure discovered by Otto [7] and try to build a Laplace–Beltrami operator. However, second-order calculus on $P_2(M)$ is delicate and a canonical Laplace–Beltrami operator is not available in full generality; see Gigli [5, Remark 5.6]. This motivates an alternative approach based on Dirichlet forms.

2.2. Dirichlet forms and symmetric Markov processes. Let $(X, \mathcal{B}, m)$ be a σ -finite measure space and consider a nonnegative definite symmetric bilinear form $\mathcal{E}$ with a dense domain $\mathcal{D}(\mathcal{E})$ on $L^2(X, m)$. It is *closed* if $\mathcal{D}(\mathcal{E})$ is complete for the norm

$$\|u\|_{\mathcal{E}_1}^2 := \|u\|_{L^2(X, m)}^2 + \mathcal{E}(u, u).$$

A *Dirichlet form* is a closed symmetric form satisfying the Markov property (stability under normal contractions), see [4, §1.4].

Closed symmetric forms correspond to non-positive self-adjoint operators: there exists a unique non-positive self-adjoint operator L on $L^2(X, m)$ such that

$$(2.3) \quad \mathcal{E}(u, v) = (-Lu, v)_{L^2(X, m)}, \quad u \in \mathcal{D}(\mathcal{E}), \quad v \in \mathcal{D}(L),$$

and conversely [4, Theorem 1.3.1]. Using the operator L , we obtain a semigroup $\{T_t\}$, and thus a Markov (or a Hunt) process. Indeed, if $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ is a *regular* Dirichlet form on a locally compact separable metric space, then there exists an m -symmetric Hunt process whose Dirichlet form is $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ [4, Theorem 7.2.1]. Such a process is unique in the sense that the transition functions agree except on the *properly exceptional sets*, i.e., sets that have zero measure under m and are avoided by the process almost surely when started outside them [4, Theorem 4.2.8]

2.3. Dirichlet–Ferguson measure. Let M be a closed Riemannian manifold with normalized volume measure m and fix a parameter $\beta > 0$.

Definition 2.1 (Poisson–Dirichlet distribution). The (one-parameter) Poisson–Dirichlet distribution $\text{PD}(\beta)$ is a probability law on decreasing sequences $s = (s_i)_{i \geq 1}$ with $s_i \geq 0$ and $\sum_{i \geq 1} s_i = 1$. One way to construct it is by *stick-breaking*: take i.i.d.

$$V_1, V_2, \dots \sim \text{Beta}(1, \beta),$$

set

$$(2.4) \quad \tilde{s}_1 := V_1, \quad \tilde{s}_n := V_n \prod_{j=1}^{n-1} (1 - V_j), \quad n \geq 2,$$

and then rank the sequence $(\tilde{s}_n)_{n \geq 1}$ in nonincreasing order.

Remark 2.2. This construction is said to be stick-breaking because one can think of starting with a stick of length 1 and breaking off a fraction V_1 to form $\tilde{s}_1$; then breaking off a fraction V_2 of the remaining piece to form $\tilde{s}_2$, and so on.

Definition 2.3 (Dirichlet–Ferguson measure). The *Dirichlet–Ferguson measure* (or *Dirichlet process*) with base measure m and concentration parameter β is the law on $P(M)$ of the random probability measure

$$(2.5) \quad \mu := \sum_{i=1}^{\infty} s_i \delta_{X_i},$$

where $(s_i)_{i \geq 1} \sim \text{PD}(\beta)$ and $(X_i)_{i \geq 1}$ are i.i.d. samples from m , independent of (s_i) . We denote this law by $\mathcal{D}_m$.

By construction, $\mathcal{D}_m$ -a.e. measure is purely atomic. In particular, $\mathcal{D}_m$ is supported on a dense subset of $P_2(M)$ [2, p. 596], and therefore has full support. In what follows we will mostly work on $M = \mathbb{T}^d$ equipped with its normalized volume m ; in that case we simply write $\mathcal{D}$ for $\mathcal{D}_m$.

2.4. Dirichlet–Ferguson diffusion and particle representation. Dello Schiavo [2] constructs a $\mathcal{D}_m$ -symmetric Markov process on $P(M)$ by Dirichlet form methods. We will call this process the *Dirichlet–Ferguson diffusion*. A key feature (and the main reason the Dirichlet–Ferguson measure is convenient here) is the availability of a particle representation. To be more specific, the Dirichlet–Ferguson diffusion on the Wasserstein space on $\mathbb{T}^d$ can be seen as a countable superposition of independent Brownian motions on $\mathbb{T}^d$. Before introducing it formally, we need some notation. Let

$$\mathbf{T}_o^\infty := \left\{ s = (s_i)_{i \geq 1} : s_1 > s_2 > \dots > 0, \sum_{i \geq 1} s_i = 1 \right\},$$

and let $(\mathbb{T}^d)_o^\infty$ be the set of sequences $x = (x_i)_{i \geq 1}$ in $\mathbb{T}^d$ with pairwise distinct coordinates. Define the “embedding” map

$$(2.6) \quad \text{em}(s, x) := \sum_{i \geq 1} s_i \delta_{x_i} \in P(\mathbb{T}^d).$$

Since under $\mathcal{D}$ the weights are almost surely strictly decreasing and the atoms are almost surely distinct, the image of em is a Borel set $O \subset P(\mathbb{T}^d)$ with $\mathcal{D}(O) = 1$ such that em admits a measurable right-inverse

$$\text{em}^{-1} : O \rightarrow \mathbf{T}_o^\infty \times (\mathbb{T}^d)_o^\infty, \quad \text{em}(\text{em}^{-1}(\mu)) = \mu.$$

(See [1, §2.3] for a concrete construction.)

2.4.1. Canonical probability space. Although the Dirichlet–Ferguson diffusion can be constructed abstractly via Dirichlet forms and thus do not depend on the underlying probability space, the theory we borrow from Delarue–Martini–Sodini [1] relies on a specific realization of the process. For this, we will first construct the underlying probability space which we call the *canonical probability space* for the Dirichlet–Ferguson diffusion.

Fix a horizon $T > 0$. Following [1, §2.3.1], we work on the space

$$\Xi_0 := \mathbf{T}_o^\infty \times (\mathbb{T}^d)_o^\infty \times (C([0, T]; \mathbb{R}^d))^\infty,$$

with coordinates $(\varsigma, \xi_0, \beta) = ((\varsigma_i)_{i \geq 1}, (\xi_{0,i})_{i \geq 1}, (\beta_i)_{i \geq 1})$. These coordinates represent the initial data of the diffusion and the family of Brownian motions driving the particles. We then set

$$\Xi := \Xi_0 \times (C([0, T]; \mathbb{R}^d))^\infty$$

and denote by $\xi = (\xi_i)_{i \geq 1}$ the last coordinate which records the particle trajectories. Let $(\mathcal{G}_t)_{t \in [0, T]}$ be the canonical filtration generated by these coordinates.

For each $t_0 \in [0, T]$ and each “mass–position” pair $(s, x) \in \mathbf{T}_o^\infty \times (\mathbb{T}^d)_o^\infty$, [1, eq. (2.29)] defines a probability measure $\mathbb{P}^{t_0, (s, x)}$ on $(\Xi, \mathcal{G}_T)$ such that under $\mathbb{P}^{t_0, (s, x)}$ the coordinates satisfy

$$(2.7) \quad \varsigma \equiv s, \quad \xi_0 \equiv x, \quad \xi_i(t) = x_i + \sqrt{\frac{2}{s_i}}(\beta_i(t) - \beta_i(t_0)), \quad t \in [t_0, T], \quad i \geq 1,$$

and $(\beta_i)_{i \geq 1}$ are independent $\mathbb{R}^d$ -Brownian motions. (As usual we interpret $\xi_i(t)$ as a $\mathbb{T}^d$ -valued process via projection modulo $\mathbb{Z}^d$.) The associated measure-valued process is

$$(2.8) \quad \mu_t := \sum_{i \geq 1} \varsigma_i \delta_{\xi_i(t)} \in P(\mathbb{T}^d), \quad t \in [t_0, T].$$

When $\mu_0 \in O$ and $(s, x) = \text{em}^{-1}(\mu_0)$, we write

$$\mathbb{P}^{\text{em}^{-1}(\mu_0)} := \mathbb{P}^{0, (s, x)}.$$

More generally, for any $\tau \in [0, T]$ and any $\mu \in O$, if $(\varsigma, x) = \text{em}^{-1}(\mu)$, we write

$$\mathbb{P}^{\tau, \text{em}^{-1}(\mu)} := \mathbb{P}^{\tau, (\varsigma, x)}.$$

In particular, the notation $\mathbb{P}^{s, \text{em}^{-1}(\mu)}$ used later means the law of the free particle system started from μ at time s . Under $\mathbb{P}^{\text{em}^{-1}(\mu_0)}$, the process $(\mu_t)_{t \in [0, T]}$ is the Dirichlet–Ferguson diffusion started from μ_0 [1, §2.3].

2.5. Cylinder functions, Wasserstein gradients and Wasserstein–Sobolev spaces. We briefly recall the functional-analytic framework of [1].

2.5.1. Smooth cylinder functions. A basic class of test functions on $P(\mathbb{T}^d)$ are *smooth cylinder functions* of the form

$$(2.9) \quad u(\mu) = F(\mu(f_1), \dots, \mu(f_k)),$$

where $k \in \mathbb{N}$, $F \in C_b^\infty(\mathbb{R}^k)$, $f_1, \dots, f_k \in C^\infty(\mathbb{T}^d)$ and $\mu(f) := \int f d\mu$. (For the Dirichlet–Ferguson measure one needs a slightly richer class allowing also “mass perturbations”; see [1, §2.2]. For the present note, (2.9) is sufficient to explain the main ideas.)

2.5.2. Wasserstein gradient on cylinder functions. For a cylinder function (2.9), the Otto calculus suggests that its gradient at μ is an element of the tangent space $T_\mu P_2(\mathbb{T}^d) \simeq L^2(\mathbb{T}^d, \mu; \mathbb{R}^d)$ and is given pointwise by

$$(2.10) \quad (Du)(\mu, x) := (\nabla u)_\mu(x) = \sum_{j=1}^k \partial_j F(\mu(f_1), \dots, \mu(f_k)) \nabla f_j(x),$$

see [1, eq. (2.11)].

2.5.3. *Cheeger energy and $H^{1,2}$.* Let $H := L^2(P(\mathbb{T}^d), \mathcal{D})$. The Cheeger energy $\text{CE} : H \rightarrow [0, \infty]$ is defined as the $L^2(\mathcal{D})$ -lower semicontinuous relaxation of the squared local Lipschitz constant with respect to W_2 [1, Definition 2.9]. The metric Sobolev space is

$$H^{1,2} := H^{1,2}(P(\mathbb{T}^d), W_2, \mathcal{D}) := \{u \in H : \text{CE}(u) < \infty\},$$

endowed with the norm

$$(2.11) \quad \|u\|_{H^{1,2}}^2 := \|u\|_H^2 + \text{CE}(u).$$

2.5.4. *Gradient for $H^{1,2}$ functions.* Let

$$\mathcal{D}(\text{d}\mu, \text{d}x) := (\delta_\mu \otimes \mu)(\text{d}\mu, \text{d}x) \mathcal{D}(\text{d}\mu)$$

be the “lift” of $\mathcal{D}$ to $P(\mathbb{T}^d) \times \mathbb{T}^d$. Delarue–Martini–Sodini prove that for each $u \in H^{1,2}$ there exists a unique

$$Du \in L^2(P(\mathbb{T}^d) \times \mathbb{T}^d, \mathcal{D}; \mathbb{R}^d)$$

characterized by approximation by cylinder functions and such that

$$(2.12) \quad \text{CE}(u, v) = \int_{P(\mathbb{T}^d) \times \mathbb{T}^d} \langle Du(\mu, x), Dv(\mu, x) \rangle \mathcal{D}(\text{d}\mu, \text{d}x), \quad u, v \in H^{1,2},$$

see [1, Proposition 2.13].

2.5.5. *Dirichlet–Ferguson Laplacian.* The bilinear form $(u, v) \mapsto \text{CE}(u, v)$ on $H^{1,2}$ is a Dirichlet form on H [1, Theorem 2.11]. Its generator is the *Dirichlet–Ferguson Laplacian* Δ with domain

$$D(\Delta) := \{v \in H^{1,2} : \exists \Delta v \in H \text{ such that } \text{CE}(u, v) = -(u, \Delta v)_H \ \forall u \in H^{1,2}\},$$

compare [1, eq. (2.24)].

2.5.6. *Chain rule.* We will repeatedly use the usual chain rule for Sobolev functions.

Proposition 2.4 (Chain rule, Proposition 5.2(e), [3]). *If $u \in H^{1,2}$ and $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is C^1 with bounded derivative, then $\phi \circ u \in H^{1,2}$ and*

$$(2.13) \quad D(\phi \circ u) = \phi'(u) Du, \quad \mathcal{D}\text{-a.e.}$$

In particular, if $u > 0$ and $\log$ is applied on a range where it is Lipschitz, then $D \log u = Du/u$.

3. MAIN RESULTS

We keep the notation of [1]. Set

$$(3.1) \quad H := L^2(P(\mathbb{T}^d), \mathcal{D}), \quad H^{1,2} := H^{1,2}(P(\mathbb{T}^d), W_2, \mathcal{D}).$$

Theorem 3.1 (Heat equation and martingale representation). *Let $g \in H$ and $T > 0$. There exists a unique*

$$u \in C([0, T]; H) \cap \text{AC}_{\text{loc}}((0, T]; H) \cap L^2([0, T]; H^{1,2})$$

such that $u_t \in D(\Delta)$ for a.e. $t \in (0, T)$ and

$$(3.2) \quad \begin{cases} \partial_t u_t + \Delta u_t = 0, & \text{for a.e. } t \in (0, T), \\ u_T = g. \end{cases}$$

Moreover, for any initial condition $\mu_0 \in \mathcal{O}$ and the Dirichlet–Ferguson diffusion $(\mu_t)_{t \in [0, T]}$ constructed on $(\Xi, \mathcal{G}_T, \mathbb{P}^{\text{em}^{-1}(\mu_0)})$ by (2.8), we have the martingale representation

$$(3.3) \quad u_t(\mu_t) = u_0(\mu_0) + \sum_{i=1}^{\infty} \int_0^t \sqrt{2\varsigma_i} Du_r(\mu_r, \xi_i(r)) \cdot \text{d}\beta_i(r), \quad t \in [0, T], \quad \mathbb{P}^{\text{em}^{-1}(\mu_0)}\text{-a.s.}$$

where $(\beta_i)_{i \geq 1}$ are the coordinate Brownian motions on Ξ .

Proof. The well-posedness of (3.2) is Corollary 3.5 in [1] (for the case $b = 0$ and $f = 0$). For the martingale representation, Proposition 4.21 in [1] yields, for $0 \leq t \leq T$,

$$g(\mu_T) = u_t(\mu_t) + \sum_{i=1}^{\infty} \int_t^T \sqrt{2\varsigma_i} Du_r(\mu_r, \xi_i(r)) \cdot d\beta_i(r), \quad \mathbb{P}^{\text{em}^{-1}(\mu_0)}\text{-a.s.}$$

Evaluating at $t = 0$ and a general t and subtracting the two identities gives (3.3). $\square$

3.1. A Doob h-transform via Girsanov. To define $\log u_t(\mu_t)$ we assume strict positivity.

Assumption 3.2. The terminal datum $g \in H \cap L^\infty(\mathcal{D})$ is bounded and satisfies $g \geq c > 0$.

We fix, for each $t \in [0, T]$, the Borel representative of u_t furnished by Theorem 4.19 in [1]. For this representative, the martingale formula in Theorem 3.1 yields

$$u_t(\mu_t) = \mathbb{E}^{\mathbb{P}^{\text{em}^{-1}(\mu_0)}}[g(\mu_T) \mid \mathcal{G}_t] \geq c, \quad t \in [0, T], \quad \mathbb{P}^{\text{em}^{-1}(\mu_0)}\text{-a.s.}$$

In particular, the pathwise quantity $u_t(\mu_t)$ is well defined for the chosen Borel representative and is strictly positive for all $t \in [0, T]$.

Let

$$h_t := u_t(\mu_t), \quad Y_t := \log h_t.$$

By (3.3), $(h_t)_{t \in [0, T]}$ is a continuous square-integrable martingale under $\mathbb{P}^{\text{em}^{-1}(\mu_0)}$. More precisely,

$$(3.4) \quad dh_t = \sum_{i=1}^{\infty} \sqrt{2\varsigma_i} Du_t(\mu_t, \xi_i(t)) \cdot d\beta_i(t),$$

and its quadratic variation is

$$(3.5) \quad d[h]_t = \sum_{i=1}^{\infty} 2\varsigma_i \|Du_t(\mu_t, \xi_i(t))\|^2 dt.$$

Indeed, the cross-variations vanish because $(\beta_i)_{i \geq 1}$ have independent components, and the series is summable thanks to [1, eq. (4.54)]. **Check**

Applying Itô's formula to the real semimartingale h_t and the function $\log$ yields

$$(3.6) \quad dY_t = \frac{1}{h_t} dh_t - \frac{1}{2h_t^2} d[h]_t.$$

Using (3.5) and the chain rule (2.13) (so that $D \log u_t = Du_t/u_t$), we obtain

$$(3.7) \quad dY_t = \sum_{i=1}^{\infty} \sqrt{2\varsigma_i} D \log u_t(\mu_t, \xi_i(t)) \cdot d\beta_i(t) - \sum_{i=1}^{\infty} \varsigma_i \|D \log u_t(\mu_t, \xi_i(t))\|^2 dt.$$

Set

$$Z_t := \exp(Y_t - Y_0) = \frac{u_t(\mu_t)}{u_0(\mu_0)}.$$

Since (h_t) is a positive martingale and μ_0 is deterministic, $\mathbb{E}^{\mathbb{P}^{\text{em}^{-1}(\mu_0)}}[Z_t] = 1$. Writing $\beta_t := (\beta_i(t))_{i \geq 1}$ and viewing it as a Brownian motion in the Hilbert space $\ell^2(\mathbb{N}; \mathbb{R}^d)$, (3.7) shows that Z is the Doléans–Dade exponential of the stochastic integral with integrand

$$(3.8) \quad \sigma_t := (\sqrt{2\varsigma_1} D \log u_t(\mu_t, \xi_1(t)), \sqrt{2\varsigma_2} D \log u_t(\mu_t, \xi_2(t)), \dots) \in \ell^2(\mathbb{N}; \mathbb{R}^d).$$

Define a new probability measure $\tilde{\mathbb{P}}$ on $(\Xi, \mathcal{G}_T)$ by

$$(3.9) \quad \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}^{\text{em}^{-1}(\mu_0)}} \Big|_{\mathcal{G}_T} = Z_T = \frac{g(\mu_T)}{u_0(\mu_0)}.$$

By Girsanov's theorem in Hilbert spaces [6, Theorem I.0.6], the processes

$$(3.10) \quad \widehat{\beta}_i(t) := \beta_i(t) - \int_0^t \sqrt{2\varsigma_i} D \log u_r(\mu_r, \xi_i(r)) dr$$

are independent Brownian motions under $\widetilde{\mathbb{P}}$. Consequently, under $\widetilde{\mathbb{P}}$ each particle satisfies the SDE

$$(3.11) \quad d\xi_i(t) = \sqrt{\frac{2}{\varsigma_i}} d\widehat{\beta}_i(t) + 2 D \log u_t(\mu_t, \xi_i(t)) dt,$$

and the induced measure-valued process $\mu_t = \sum_{i \geq 1} \varsigma_i \delta_{\xi_i(t)}$ solves a drifted version of the Dirichlet-Ferguson diffusion. The change of measure (3.9) can be interpreted as a Doob h -transform (or conditioning by terminal weight) since it reweights paths according to $g(\mu_T)$.

3.2. Finite dimensional approximation. We will now consider the finite dimensional approximation of the diffusion with drift constructed in the previous subsection. Our main goal is to complete the following program:

- (1) Identify the transition family associated with the drifted process.
- (2) Describe the finite-dimensional approximation of the drifted process and identify its transition family.
- (3) Prove that the finite-dimensional transition families converge to the infinite-dimensional one.

3.2.1. The transition family of the drifted process. More precisely, since the drifted dynamics are time-inhomogeneous, the natural object is a two-parameter transition family. For $0 \leq s \leq T$ and $\mu \in O$, let $\widetilde{\mathbb{P}}^{s,\mu}$ be the probability measure on $(\Xi, \mathcal{G}_T)$ defined by

$$(3.12) \quad \frac{d\widetilde{\mathbb{P}}^{s,\mu}}{d\mathbb{P}^{s,\text{em}^{-1}(\mu)}} \Big|_{\mathcal{G}_T} = \frac{g(\mu_T)}{u_s(\mu)}.$$

When $s = 0$ and $\mu = \mu_0$, this coincides with the measure $\widetilde{\mathbb{P}}$ introduced above. Under $\widetilde{\mathbb{P}}^{s,\mu}$, the coordinate process solves the drifted particle system

$$(3.13) \quad \mu_t := \sum_{i \geq 1} \varsigma_i \delta_{\xi_i(t)}, \quad d\xi_i(t) = \sqrt{\frac{2}{\varsigma_i}} d\widehat{\beta}_i(t) + 2 D \log u_t(\mu_t, \xi_i(t)) dt, \quad t \in [s, T].$$

We define its transition family by

$$(\widetilde{P}_{s,t}f)(\mu) := \mathbb{E}^{\widetilde{\mathbb{P}}^{s,\mu}}[f(\mu_t)], \quad 0 \leq s \leq t \leq T.$$

Proposition 3.3. *For every $0 \leq s \leq t \leq T$ and $f \in H$,*

$$(3.14) \quad (\widetilde{P}_{s,t}f)(\mu) = \frac{1}{u_s(\mu)} P_{t-s}(u_t f)(\mu).$$

Proof. By the definition of $\widetilde{\mathbb{P}}^{s,\mu}$,

$$\mathbb{E}^{\widetilde{\mathbb{P}}^{s,\mu}}[f(\mu_t)] = \frac{1}{u_s(\mu)} \mathbb{E}^{\mathbb{P}^{s,\text{em}^{-1}(\mu)}}[f(\mu_t)g(\mu_T)].$$

Using the Markov property at time t and the Borel representative of u_t fixed above, we obtain

$$\mathbb{E}^{\mathbb{P}^{s,\text{em}^{-1}(\mu)}}[g(\mu_T) \mid \mathcal{G}_t] = u_t(\mu_t),$$

hence

$$\mathbb{E}^{\mathbb{P}^{s,\text{em}^{-1}(\mu)}}[f(\mu_t)g(\mu_T)] = \mathbb{E}^{\mathbb{P}^{s,\text{em}^{-1}(\mu)}}[f(\mu_t)u_t(\mu_t)] = P_{t-s}(u_t f)(\mu).$$

This proves the claim. $\square$

Remark 3.4. The expression $P_{t-s}(u_t f)$ is well defined for each $f \in H$ because $u_t = P_{T-t}g$ is bounded by $\|g\|_{L^\infty}$.

3.2.2. *Finite-dimensional approximation of the free process.* We now recall the finite-dimensional approximation used in [2, Section 6.1]. Let

$$T := \left\{ s = (s_i)_{i \geq 1} \in [0, 1]^{\mathbb{N}} : \sum_{i=1}^{\infty} s_i = 1, s_1 \geq s_2 \geq \dots \geq 0 \right\},$$

and let

$$T^\circ := \left\{ s \in T : s_1 > s_2 > \dots > 0 \right\}.$$

For each $n \in \mathbb{N}$, define the coordinate projection

$$\text{pr}_n : T \rightarrow \mathbb{R}^n, \quad \text{pr}_n(s_1, s_2, \dots) = (s_1, \dots, s_n).$$

We now define

$$\Sigma_n := \text{pr}_n(T), \quad \Sigma_n^{1/n} := \{s = (s_1, \dots, s_n) \in \Sigma_n : s_1 \geq 1/n\},$$

and define

$$\widehat{M}_n := \Sigma_n^{1/n} \times (\mathbb{T}^d)^n.$$

We endow $\Sigma_n^{1/n}$ with the measure

$$\pi_n^\beta := (\text{pr}_n)_\# \text{PD}(\beta)|_{\Sigma_n^{1/n}},$$

and $\widehat{M}_n$ with the product measure

$$\widehat{m}_n := \pi_n^\beta \otimes m^{\otimes n}.$$

We also define the map

$$\Phi_n : \widehat{M}_n \rightarrow \mathcal{P}(\mathbb{T}^d), \quad \Phi_n(s, x) := \sum_{i=1}^n s_i \delta_{x_i}.$$

For each fixed $s \in \Sigma_n^{1/n}$, let

$$W_t^{n,s} = (\xi^1(t), \dots, \xi^n(t)), d\xi^i(t) = \sqrt{\frac{2}{s_i}} d\beta_i(t), \quad t \in [0, T], \quad i = 1, \dots, n,$$

where $\beta_1, \dots, \beta_n$ are independent Brownian motions on $\mathbb{T}^d$. Equivalently, the generator on the fiber $(\mathbb{T}^d)^n$ is

$$L^{(n),s} = \sum_{i=1}^n s_i^{-1} \Delta_{x_i}.$$

The corresponding measure-valued process is

$$\mu_t^{(n)} = \Phi_n(s, W_t^{n,s}) = \sum_{i=1}^n s_i \delta_{\xi_t^i}.$$

We will consider the Hilbert space for the finite dimensional approximation to be

$$H_n := L^2(\widehat{M}_n, \widehat{m}_n).$$

The free n -particle system naturally lives on $\widehat{M}_n$: the mass vector is frozen and only the position variable evolves. Accordingly, if $f \in H_n$ and $(r, x) \in \widehat{M}_n$, we define

$$(3.15) \quad (P_t^{(n)} f)(r, x) := \mathbb{E}[f(r, W_t^{n,r}) \mid W_0^{n,r} = x].$$

Its generator acts only on the torus variables and is given by

$$L^{(n)} = \sum_{i=1}^n r_i^{-1} \Delta_{x_i}.$$

CHECK For a function $g_n \in H_n$ with $g_n \geq c > 0$, let $u_t^{(n)}$ be the solution to the finite-dimensional backward heat equation

$$(3.16) \quad \begin{cases} \partial_t u_t^{(n)} + L^{(n)} u_t^{(n)} = 0, & t \in (0, T), \\ u_T^{(n)} = g_n. \end{cases}$$

To do calculus on $\widehat{M}_n$, we keep the mass variable fixed and differentiate only in the torus directions. In exact analogy with the infinite-dimensional construction, the gradient must therefore be defined first on smooth functions and then extended by closure to the corresponding Sobolev space.

We introduce the weighted gradient space

$$\mathfrak{H}_n := L^2(\widehat{M}_n, \widehat{m}_n; (\mathbb{R}^d)^n), \quad \|V\|_{\mathfrak{H}_n}^2 := \int_{\widehat{M}_n} \sum_{i=1}^n r_i |V_i(r, x)|^2 \widehat{m}_n(dr, dx).$$

For $u \in C^1(\widehat{M}_n)$ we define the smooth fiber-gradient

$$(3.17) \quad D_{\text{sm}}^{(n)} u(r, x) := \left(\frac{1}{r_1} \nabla_{x_1} u(r, x), \dots, \frac{1}{r_n} \nabla_{x_n} u(r, x) \right),$$

where the values on the $\widehat{m}_n$ -negligible set $\{r_i = 0\}$ are irrelevant. The associated quadratic form is

$$(3.18) \quad \mathcal{E}_n^{\text{sm}}(u, v) := \int_{\widehat{M}_n} \sum_{i=1}^n r_i \left\langle D_{\text{sm},i}^{(n)} u(r, x), D_{\text{sm},i}^{(n)} v(r, x) \right\rangle \widehat{m}_n(dr, dx) = \int_{\widehat{M}_n} \sum_{i=1}^n \frac{1}{r_i} \langle \nabla_{x_i} u(r, x), \nabla_{x_i} v(r, x) \rangle \widehat{m}_n(dr, dx).$$

This is the finite-dimensional counterpart of the identity (2.12).

Proposition 3.5 (Finite-dimensional Sobolev space and gradient). *The form $(\mathcal{E}_n^{\text{sm}}, C^1(\widehat{M}_n))$ is closable on H_n . We denote its closure by $(\mathcal{E}_n, H_n^{1,2})$ and endow $H_n^{1,2}$ with the norm*

$$\|u\|_{H_n^{1,2}}^2 := \|u\|_{H_n}^2 + \mathcal{E}_n(u, u).$$

The operator $D_{\text{sm}}^{(n)} : C^1(\widehat{M}_n) \rightarrow \mathfrak{H}_n$ is closable. Its closure, still denoted by

$$D^{(n)} : H_n^{1,2} \longrightarrow \mathfrak{H}_n,$$

is characterized as follows: for every $u \in H_n^{1,2}$ there exists a sequence $u_k \in C^1(\widehat{M}_n)$ such that

$$u_k \rightarrow u \text{ in } H_n, \quad D_{\text{sm}}^{(n)} u_k \rightarrow D^{(n)} u \text{ in } \mathfrak{H}_n,$$

and the limit $D^{(n)} u$ does not depend on the approximating sequence. Moreover,

$$(3.19) \quad \mathcal{E}_n(u, v) = \left\langle D^{(n)} u, D^{(n)} v \right\rangle_{\mathfrak{H}_n}, \quad u, v \in H_n^{1,2}.$$

Proof. The closability of $(\mathcal{E}_n^{\text{sm}}, C^1(\widehat{M}_n))$ is precisely Proposition 6.2 in [2]. On $C^1(\widehat{M}_n)$ we have $\mathcal{E}_n^{\text{sm}}(u) = \|D_{\text{sm}}^{(n)} u\|_{\mathfrak{H}_n}^2$. Hence, if $u_k \in C^1(\widehat{M}_n)$ satisfies $u_k \rightarrow 0$ in H_n and $D_{\text{sm}}^{(n)} u_k \rightarrow V$ in $\mathfrak{H}_n$, then (u_k) is $\mathcal{E}_n^{\text{sm}}$ -Cauchy; by closability, $\mathcal{E}_n^{\text{sm}}(u_k, u_k) \rightarrow 0$, so $\|V\|_{\mathfrak{H}_n} = 0$. Therefore $D_{\text{sm}}^{(n)}$ is closable. The uniqueness of the limit follows from the standard argument of the closability. Finally, the identity (3.19) follows from the polarization from the equality $\mathcal{E}_n(u, u) = \|D^{(n)} u\|_{\mathfrak{H}_n}^2$. $\square$

Now fix $(s, x) \in \widehat{M}_n$ and define

$$h_t^{(n)} := u_t^{(n)}(s, W_t^{n,s}).$$

This process satisfies the following properties.

Proposition 3.6. (1) $h_t^{(n)} \geq c$.

(2) $h_t^{(n)}$ is a martingale and satisfies

$$(3.20) \quad dh_t^{(n)} = \sum_{i=1}^n \sqrt{2s_i} D_i^{(n)} u_t^{(n)}(s, W_t^{n,s}) \cdot d\beta_i(t).$$

Proof. For the lower bound, use the positivity preserving property of the semigroup $P_t^{(n)}$:

$$u_t^{(n)} = P_{T-t}^{(n)} g_n \geq c P_{T-t}^{(n)} \mathbf{1} = c,$$

hence $h_t^{(n)} = u_t^{(n)}(s, W_t^{n,s}) \geq c$ for every $t \in [0, T]$.

For the martingale statement, fix the mass vector s and write

$$v_n(t, z) := u_t^{(n)}(s, z), \quad z = (z_1, \dots, z_n) \in (\mathbb{T}^d)^n.$$

Then v_n solves

$$\partial_t v_n + \sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_n = 0.$$

Applying Itô's formula to $t \mapsto v_n(t, W_t^{n,s}) = h_t^{(n)}$ gives Check ito and smoothness. gradient doesnt work for non smooth vn

$$\begin{aligned} dh_t^{(n)} &= \left(\partial_t v_n + \sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_n \right) (t, W_t^{n,s}) dt + \sum_{i=1}^n \sqrt{\frac{2}{s_i}} \nabla_{x_i} v_n(t, W_t^{n,s}) \cdot d\beta_i(t) \\ &= \sum_{i=1}^n \sqrt{\frac{2}{s_i}} \nabla_{x_i} u_t^{(n)}(s, W_t^{n,s}) \cdot d\beta_i(t) \\ &= \sum_{i=1}^n \sqrt{2s_i} D_i^{(n)} u_t^{(n)}(s, W_t^{n,s}) \cdot d\beta_i(t). \end{aligned}$$

Hence $h_t^{(n)}$ is a local martingale. The uniform lower bound and boundedness of $u_t^{(n)}$ imply square-integrability on every finite time interval, so the local martingale is in fact a martingale.

If one starts from merely bounded terminal data g_n , the same identity follows by approximating g_n by smooth terminal conditions and passing to the limit; we omit this standard regularization argument. □

By following the same argument as in the infinite-dimensional case, the density process for the finite-dimensional h -transform is built from

$$Z_t^{(n)} := \frac{u_t^{(n)}(s, W_t^{n,s})}{u_0^{(n)}(s, x)},$$

and the corresponding Girsanov shift is

$$(3.21) \quad \widehat{\beta}_i^{(n)}(t) = \beta_i(t) - \int_0^t \sqrt{2s_i} D_i^{(n)} \log u_r^{(n)}(s, W_r^{n,s}) dr.$$

Consequently,

$$(3.22) \quad d\xi_i^{(n)}(t) = \sqrt{\frac{2}{s_i}} d\widehat{\beta}_i^{(n)}(t) + 2D_i^{(n)} \log u_t^{(n)}(s, W_t^{n,s}) dt.$$

3.2.3. *The transition family of the finite-dimensional approximation and its convergence.* Similar to the infinite-dimensional case, for $0 \leq \tau \leq t \leq T$ and $(r, x) \in \widehat{M}_n$ let $\mathbb{P}^{(n), \tau, (r, x)}$ denote the law of the free finite-dimensional particle system started from (r, x) at time τ , and let $\widetilde{\mathbb{P}}^{(n), \tau, (r, x)}$ be its h -transform defined by

$$\frac{d\widetilde{\mathbb{P}}^{(n), \tau, (r, x)}}{d\mathbb{P}^{(n), \tau, (r, x)}} = \frac{g_n(r, W_T^{n, r})}{u_\tau^{(n)}(r, x)}.$$

We then set

$$(\tilde{P}_{\tau, t}^{(n)} f)(r, x) := \mathbb{E}^{\widetilde{\mathbb{P}}^{(n), \tau, (r, x)}} [f(r, W_t^{n, r})].$$

Proposition 3.7. *For every $0 \leq \tau \leq t \leq T$,*

$$(3.23) \quad (\tilde{P}_{\tau, t}^{(n)} f)(r, x) = \frac{1}{u_\tau^{(n)}(r, x)} P_{t-\tau}^{(n)}(u_t^{(n)} f)(r, x).$$

Proof. The proof is the same Bayes/Markov argument as in the infinite-dimensional case. $\square$

Finally, we will show that $\tilde{P}_{s, t}^{(n)}$ converges to $\tilde{P}_{s, t}$ as $n \rightarrow \infty$. We will first define the mode of the convergence.

Definition 3.8 (Generalized Hilbert convergence). Following [2, Appendix A.3], let H and $(H_n)_{n \in \mathbb{N}}$ be Hilbert spaces and let $\mathcal{H} = H \coprod \bigsqcup_{n \in \mathbb{N}} H_n$. Let $D \subset H$ be dense, and let $\Lambda_n : D \rightarrow H_n$ be densely defined linear maps. We say that H_n $\mathcal{H}$ -converges to H if

$$\|\Lambda_n u\|_{H_n} \xrightarrow{n \rightarrow \infty} \|u\|_H \quad \text{for all } u \in D.$$

A sequence $u_n \in H_n$ is said to $\mathcal{H}$ -strongly converge to $u \in H$ if there exists a sequence $\tilde{u}^m \in D$ such that

$$\|\tilde{u}^m - u\|_H \xrightarrow{m \rightarrow \infty} 0, \quad \lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \|\Lambda_n \tilde{u}^m - u_n\|_{H_n} = 0.$$

A sequence $u_n \in H_n$ is said to $\mathcal{H}$ -weakly converge to $u \in H$ if for every sequence $v_n \in H_n$ that $\mathcal{H}$ -strongly converges to some $v \in H$ one has

$$\langle u_n, v_n \rangle_{H_n} \xrightarrow{n \rightarrow \infty} \langle u, v \rangle_H.$$

Let $B_n \in \mathcal{B}(H_n)$ be bounded operators and let $B \in \mathcal{B}(H)$. We say that B_n converges $\mathcal{H}$ -strongly to B if for every sequence $u_n \in H_n$ that $\mathcal{H}$ -strongly converges to $u \in H$, the sequence $B_n u_n$ $\mathcal{H}$ -strongly converges to Bu .

We now map this definition into our setting. Our limiting Hilbert space is $H = L^2(P(\mathbb{T}^d), \mathcal{D})$, while the approximating Hilbert spaces are the previously defined spaces $H_n = L^2(\widehat{M}_n, \widehat{m}_n)$. We will prove a convenient representation of H_n as a closed subspace of H and identify the corresponding orthogonal projection. For this, we introduce the following class of functions on $P(\mathbb{T}^d)$: for $\hat{f} \in C_c^\infty(\mathbb{T}^d \times (1/n, 1])$, define

$$\hat{f}^*(\eta) := \int_{\mathbb{T}^d} \hat{f}(x, \eta(\{x\})) \eta(dx),$$

and let $\mathcal{B}_{1/n}$ be the σ -algebra generated by the cylinder functions

$$F(\hat{f}_1^*, \dots, \hat{f}_k^*), \quad F \in C_b^\infty(\mathbb{R}^k), \quad \hat{f}_j \in C_c^\infty(\mathbb{T}^d \times (1/n, 1]).$$

We also set

$$H_{1/n} := L^2(P(\mathbb{T}^d), \mathcal{B}_{1/n}, \mathcal{D}), \quad P_n := \mathbb{E}^{\mathcal{D}}[\cdot \mid \mathcal{B}_{1/n}].$$

Proposition 3.9 (cf. Lemma 6.4 and Proposition 6.5, [2]). *There is a unique isometric isomorphism*

$$U_n : H_{1/n} \longrightarrow H_n = L^2(\widehat{M}_n, \widehat{m}_n)$$

with the following property: if

$$u(\eta) = F(\hat{f}_1^*(\eta), \dots, \hat{f}_k^*(\eta)), \quad F \in C_b^\infty(\mathbb{R}^k), \quad \hat{f}_j \in C_c^\infty(\mathbb{T}^d \times (1/n, 1]),$$

then

$$(U_n u)(s, x) = F\left(\sum_{i=1}^n s_i \hat{f}_1(x_i, s_i), \dots, \sum_{i=1}^n s_i \hat{f}_k(x_i, s_i)\right), \quad (s, x) \in \widehat{M}_n.$$

If $J_n := U_n^{-1} : H_n \rightarrow H$, then J_n is an isometric embedding with

$$J_n(H_n) = H_{1/n} = \text{Ran}(P_n).$$

Hence the finite-dimensional space H_n is canonically realized inside H through the orthogonal projection P_n .

Let $\hat{\mathfrak{J}}_{1/n}^\infty$ denote the class of cylinder functions appearing in the statement.

Proposition 3.10 (Compatibility of finite- and infinite-dimensional gradients). *Let*

$$\mathfrak{H} := L^2(P(\mathbb{T}^d) \times \mathbb{T}^d, \mathcal{D}; \mathbb{R}^d), \quad \|F\|_{\mathfrak{H}}^2 = \int_{P(\mathbb{T}^d)} \int_{\mathbb{T}^d} |F(\eta, z)|^2 \eta(dz) \mathcal{D}(d\eta)$$

For $V = (V_1, \dots, V_n) \in \mathfrak{H}_n$, define $K_n V \in \mathfrak{H}$ by

$$(K_n V)(\eta, z) := \sum_{i=1}^n \mathbf{1}_{\{x_i(\eta)\}}(z) V_i(\text{pr}_n(s(\eta)), x_1(\eta), \dots, x_n(\eta)), \quad (\eta, z) \in O \times \mathbb{T}^d,$$

and set $K_n V(\eta, z) = 0$ on $O^c \times \mathbb{T}^d$ or $s_1(\eta) < 1/n$. Here, $s(\eta), x_i(\eta)$ are defined by $\text{em}^{-1}(\eta) = (s(\eta), x(\eta))$ with $s(\eta) = (s_1(\eta), s_2(\eta), \dots)$ and $x(\eta) = (x_1(\eta), x_2(\eta), \dots)$, and O is the domain of em^{-1} . Then K_n is an isometric embedding of $\mathfrak{H}_n$ into $\mathfrak{H}$.

Moreover, for every $u \in H_n^{1,2}$,

$$(3.24) \quad J_n u \in H^{1,2}, \quad D(J_n u) = K_n D^{(n)} u, \quad \text{CE}(J_n u) = \mathcal{E}_n(u).$$

In particular, if $u^{(n)} \in H_n^{1,2}$ and $J_n u^{(n)} \rightarrow u$ in $H^{1,2}$, then

$$(3.25) \quad K_n D^{(n)} u^{(n)} \rightarrow Du \quad \text{in } \mathfrak{H}.$$

Proof. For any $V \in \mathfrak{H}_n$, we have

$$\begin{aligned} \|K_n V\|_{\mathfrak{H}}^2 &= \int_{P(\mathbb{T}^d)} \int_{\mathbb{T}^d} |K_n V(\eta, z)|^2 \eta(dz) \mathcal{D}(d\eta) \\ &= \int_{P(\mathbb{T}^d)} \sum_{i=1}^n s_i(\eta) \left| V_i(\text{pr}_n(s(\eta)), x_1(\eta), \dots, x_n(\eta)) \right|^2 \mathcal{D}(d\eta) \\ &= \int_{\widehat{M}_n} \sum_{i=1}^n r_i |V_i(r, x)|^2 \widehat{m}_n(dr, dx) = \|V\|_{\mathfrak{H}_n}^2. \end{aligned}$$

Let $\hat{\mathfrak{J}}_{1/n}^\infty$ be the cylinder class introduced in Proposition 3.9 and fix $v \in \hat{\mathfrak{J}}_{1/n}^\infty$, say

$$v(\eta) = F(\hat{f}_1^*(\eta), \dots, \hat{f}_k^*(\eta)), \quad F \in C_b^\infty(\mathbb{R}^k), \quad \hat{f}_j \in C_c^\infty(\mathbb{T}^d \times (1/n, 1]).$$

Then $U_n v \in C^1(\widehat{M}_n)$ and

$$(U_n v)(r, x) = F\left(\sum_{i=1}^n r_i \hat{f}_1(x_i, r_i), \dots, \sum_{i=1}^n r_i \hat{f}_k(x_i, r_i)\right).$$

A direct differentiation in the torus variables yields

$$D_i^{(n)}(U_n v)(r, x) = \sum_{j=1}^k \partial_j F(\cdots) \nabla_x \hat{f}_j(x_i, r_i), \quad i = 1, \dots, n.$$

Therefore,

$$K_n D^{(n)}(U_n, v)(\eta, z) = \sum_{i=1}^n \mathbf{1}_{\{x_i(\eta)\}}(z) \sum_{j=1}^k \partial_j F(\cdots) \nabla_x \hat{f}_j(x_i(\eta), s_i(\eta)).$$

which equals

$$\sum_{j=1}^k \partial_j F(\cdots) \nabla_x \hat{f}_j(x_i, s_i(\eta))$$

if $z = x_i(\eta)$ for some i and equals 0 otherwise. This is precisely the gradient formula for cylinder functions in the infinite-dimensional setting, as given by [1, Lemma 2.8]:

$$Dv(\eta, z) = \sum_{j=1}^k \partial_j F(\cdots) \nabla_x \hat{f}_j(z, \eta(\{z\})).$$

Hence $Dv = K_n D^{(n)}(U_n v)$ and in particular $D(J_n U_n v) = K_n D^{(n)}(U_n v)$ because $J_n = U_n^{-1}$. Moreover,

$$\text{CE}(J_n U_n v) = \|Dv\|_{\mathfrak{H}}^2 = \|K_n D^{(n)}(U_n v)\|_{\mathfrak{H}}^2 = \|D^{(n)}(U_n v)\|_{\mathfrak{H}_n}^2 = \mathcal{E}_n(U_n v).$$

By Lemma 6.4 of [2] together with the density argument at the end of its proof, $U_n \hat{\mathfrak{Z}}_{1/n}^\infty$ is dense in $H_n^{1,2}$. Let now $u \in H_n^{1,2}$ and choose $u_k \in U_n \hat{\mathfrak{Z}}_{1/n}^\infty$ with $u_k \rightarrow u$ in $H_n^{1,2}$. Set $v_k := J_n u_k$. Then

$$\|v_k - v_\ell\|_{H^{1,2}}^2 = \|u_k - u_\ell\|_{H_n}^2 + \text{CE}(v_k - v_\ell) = \|u_k - u_\ell\|_{H_n}^2 + \mathcal{E}_n(u_k - u_\ell) = \|u_k - u_\ell\|_{H_n^{1,2}}^2.$$

so (v_k) is Cauchy in $H^{1,2}$. Hence by definition of closedness of the Dirichlet form, $v_k \rightarrow v$ for some $v \in H^{1,2}$ in $H^{1,2}$ and H . Since J_n is an isometry on H_n , we have $\|v_k - Ju\|_H = \|Ju_k - Ju\|_{H_n} = \|u_k - u\|_{H_n} \rightarrow 0$, so by uniqueness of limits we have $v = Ju$. Passing to the limit in

$$Dv_k = K_n D^{(n)} u_k$$

and using that both D and $D^{(n)}$ are closed while K_n is an isometry, we obtain (3.24). Finally, (3.25) is immediate: if $J_n u^{(n)} \rightarrow u$ in $H^{1,2}$, then

$$\|K_n D^{(n)} u^{(n)} - Du\|_{\mathfrak{H}} = \|D(J_n u^{(n)}) - Du\|_{\mathfrak{H}} \rightarrow 0.$$

□

From now on, whenever we compare H_n with H , we do so through the canonical embedding $J_n : H_n \rightarrow H_{1/n} \subset H$. That is, we identify the elements of H_n with their images under J_n in $H_{1/n}$. Moreover, we will always take $\Phi_n = P_n$ in the definition of $\mathcal{H}$ -convergence. With this convention, the finite-dimensional space H_n is canonically realized as a closed subspace of H through the orthogonal projection P_n . We will now turn to the main result of this section, which is the convergence of $\tilde{P}_{s,t}^{(n)}$ to $\tilde{P}_{s,t}$ in the sense of $\mathcal{H}$ -strong convergence. We remark that $P_{s,t}^{(n)}$ and $\tilde{P}_{s,t}$ involve products of functions, so we will use the following elementary lemma.

Lemma 3.11. *Let H_n, H as above and let $a_n \in H_n, a \in H, b_n \in H_n, b \in H$. Assume that*

- $a_n \rightarrow a, b_n \rightarrow b$ in the sense of $\mathcal{H}$ -strong convergence, and
- There exists a constant $C > 0$ so that $\sup_n \|a_n\|_\infty \leq C$ and $\|a\|_\infty \leq C$.

Then $a_n b_n \rightarrow ab$ in the sense of $\mathcal{H}$ -strong convergence.

Proof. Because $\mathcal{H}$ -strong convergence is defined through the maps $\Phi_n = P_n$, Lemma A.10 of [2] shows that $\mathcal{H}$ -strong convergence of a_n to a is equivalent to ordinary strong convergence of a_n to a in $H = L^2(P(\mathbb{T}^d), \mathcal{D})$, and similarly for b_n . It is therefore enough to prove that

$$\|a_n b_n - ab\|_{L^2(\mathcal{D})} \xrightarrow{n \rightarrow \infty} 0.$$

Write

$$a_n b_n - ab = a_n(b_n - b) + (a_n - a)b.$$

For the first term we simply use the uniform L^∞ bound on a_n :

$$\|a_n(b_n - b)\|_{L^2(\mathcal{D})} \leq C \|b_n - b\|_{L^2(\mathcal{D})} \xrightarrow{n \rightarrow \infty} 0.$$

For the second term, fix $M > 0$ and let

$$b^M := (-M) \vee (b \wedge M)$$

be the usual truncation of b . Then $\|b - b^M\|_{L^2(\mathcal{D})} \rightarrow 0$ as $M \rightarrow \infty$, and since $|a_n - a| \leq 2C$ almost everywhere we have

$$\begin{aligned} \|(a_n - a)b\|_{L^2(\mathcal{D})} &\leq \|(a_n - a)b^M\|_{L^2(\mathcal{D})} + \|(a_n - a)(b - b^M)\|_{L^2(\mathcal{D})} \\ &\leq M \|a_n - a\|_{L^2(\mathcal{D})} + 2C \|b - b^M\|_{L^2(\mathcal{D})}. \end{aligned}$$

First let $n \rightarrow \infty$ and use the strong convergence $a_n \rightarrow a$ in H ; then let $M \rightarrow \infty$. This shows that $\|(a_n - a)b\|_{L^2(\mathcal{D})} \rightarrow 0$.

Combining the two estimates yields $\|a_n b_n - ab\|_{L^2(\mathcal{D})} \rightarrow 0$, which is precisely the desired $\mathcal{H}$ -strong convergence of $a_n b_n$ to ab . $\square$

Theorem 3.12. $\tilde{P}_{s,t}^{(n)} : H_n \rightarrow H_n$ converges $\mathcal{H}$ -strongly to $\tilde{P}_{s,t} : H \rightarrow H$.

Proof. By Proposition 3.9, after transport by J_n the operators $P_n = \mathbb{E}^\mathcal{D}[\cdot \mid \mathcal{B}_{1/n}]$ are orthogonal projections on H with range $J_n(H_n) = H_{1/n}$, and $P_n \rightarrow \text{Id}_H$ strongly as $n \rightarrow \infty$. Hence, by [2, Lemma A.10], $\mathcal{H}$ -strong convergence is the same as strong convergence in H . We will use this repeatedly below. We follow a similar strategy as Proposition 6.5 in [2]. Let $g \in L^\infty(P(\mathbb{T}^d), \mathcal{D})$ be a function with $g \geq c > 0$ for some constant $c > 0$, and define its finite-dimensional approximation by

$$(3.26) \quad g_n := P_n g \in H_n.$$

Under this definition, we show:

- (1) That $g_n = P_n g$ converges to g in H as $n \rightarrow \infty$.
- (2) That $P^{(n)} \rightarrow P$ in the sense of $\mathcal{H}$ -strong convergence.

Once we have these two points, we have that $u_t^{(n)} = P_{T-t}^{(n)} g_n$, converges to $u_t = P_{T-t} g$ in H for each $t \in [0, T]$ by definition of the $\mathcal{H}$ -strong convergence of operators. Regarding the first point, as discussed in the proof of [2, Proposition 6.5], P_n converges strongly to Id , so $g_n = P_n g$ converges to g in H as $n \rightarrow \infty$. The second point follows from the fact that the corresponding Dirichlet forms converge in the sense of Mosco convergence, which is shown in [2, Proposition 6.5], and the Kuwae-Shioya theorem [2, Theorem A.11], which states the equivalence between Mosco convergence of Dirichlet forms and $\mathcal{H}$ -strong convergence of the associated semigroups.

Next, we will show that, for any $f_n \in H_n$ $\mathcal{H}$ -strongly converging to $f \in H$, $P_{t-s}^{(n)}(u_t^{(n)} f_n)$ converges to $P_{t-s}(u_t f)$ in H as $n \rightarrow \infty$. First, we will prove that, for any $f_n \in H_n$, $u_t^{(n)} f_n$ converges to $u_t f$ in the sense of $\mathcal{H}$ -strong convergence. This follows from Lemma 3.11 and the fact that $u_t^{(n)}$ is uniformly bounded by $\|g\|_{L^\infty}$ for each t . Then, by the $\mathcal{H}$ -strong convergence of $P_{t-s}^{(n)}$ to P_{t-s} , we have that $P_{t-s}^{(n)}(u_t^{(n)} f_n)$ converges to $P_{t-s}(u_t f)$ in H as $n \rightarrow \infty$.

Finally, we show that $\frac{1}{u_s^{(n)}}$ converges to $\frac{1}{u_s}$ in the sense of $\mathcal{H}$ -strong convergence. We first note that $g_n = P_n g$ is bounded below by c since $g \geq c$ and P_n is a positive operator. By again the positivity, we have $u_t^{(n)} = P_{T-t}^{(n)} g_n \geq c$ for each $t \in [0, T]$. Thus, $\frac{1}{u_s^{(n)}}$ is well-defined and bounded by $1/c$. Now we note that a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ defined by $\phi(x) = 1/x$ is $1/c^2$ -Lipschitz on $[c, \infty)$, so after transport to H we have

$$(3.27) \quad \left\| \frac{1}{u_s^{(n)}} - \frac{1}{u_s} \right\|_H = \|\phi(u_s^{(n)}) - \phi(u_s)\|_H \leq \frac{1}{c^2} \|u_s^{(n)} - u_s\|_H \rightarrow 0$$

as $n \rightarrow \infty$.

We will now combine the above results to show that $\tilde{P}_{s,t}^{(n)} f$ converges to $\tilde{P}_{s,t} f$ in the sense of $\mathcal{H}$ -strong convergence. This follows from the fact that $\frac{1}{u_s^{(n)}}$ and $\frac{1}{u_s}$ are uniformly bounded by $1/c$ for each n and Lemma 3.11 applied to the product of $\frac{1}{u_s^{(n)}}$ and $P_{t-s}^{(n)}(u_t^{(n)} f_n)$. $\square$

3.3. Numerical simulation. We now explain how the finite-dimensional h -transformed dynamics (3.22) can be simulated numerically once the mass vector $s \in \Sigma_n^{1/n}$ has been frozen. For a fixed s , write

$$(3.28) \quad b_i^{(n)}(t, x; s) := 2D_i^{(n)} \log u_t^{(n)}(s, x) = \frac{2}{s_i} \nabla_{x_i} \log u_t^{(n)}(s, x), \quad i = 1, \dots, n.$$

Then (3.22) becomes the time-inhomogeneous SDE

$$(3.29) \quad d\xi_i^{(n)}(t) = b_i^{(n)}(t, \xi^{(n)}(t); s) dt + \sqrt{\frac{2}{s_i}} d\hat{\beta}_i^{(n)}(t), \quad i = 1, \dots, n.$$

Since $u_t^{(n)} = P_{T-t}^{(n)} g_n$ and the semigroup acts only in the torus variables, one has the heat-kernel representation

$$(3.30) \quad u_t^{(n)}(s, x) = \int_{(\mathbb{T}^d)^n} g_n(s, y) \prod_{i=1}^n p_{(T-t)/s_i}^{\mathbb{T}^d}(x_i, y_i) dy,$$

where $p_{\tau}^{\mathbb{T}^d}$ denotes the heat kernel on $\mathbb{T}^d$ corresponding to the equation $\partial_{\tau} u = \Delta u$. Thus the numerical task is to evaluate $u_t^{(n)}$ and its logarithmic gradient and then to discretize (3.29).

Fix a time grid $0 = t_0 < t_1 < \dots < t_M = T$ with $\Delta t = T/M$. Writing $X^m = (X_1^m, \dots, X_n^m)$ for the numerical approximation to $\xi^{(n)}(t_m)$, the Euler–Maruyama update is

$$(3.31) \quad X_i^{m+1} = X_i^m + b_i^{(n)}(t_m, X^m; s) \Delta t + \sqrt{\frac{2\Delta t}{s_i}} Z_i^m, \quad Z_i^m \sim N(0, I_d),$$

followed by the projection of X_i^{m+1} back to $\mathbb{T}^d$ modulo $\mathbb{Z}^d$. The associated measure-valued approximation is then

$$\mu_{t_m}^{(n)} = \sum_{i=1}^n s_i \delta_{X_i^m}.$$

For a generic terminal condition g_n , the main computational cost is the evaluation of (3.30). This can be done by methods such as Fourier methods on the torus, or Monte Carlo evaluation of the free particle semigroup. The scheme becomes especially simple when g_n is chosen so that $u_t^{(n)}$ is explicit.

Gaussian terminal condition centred at a prescribed configuration. Because the finite-dimensional dynamics only move the particle locations, the natural way to promote the terminal measure toward a prescribed atomic target is to keep the mass vector s fixed and choose a terminal

weight concentrated around a target configuration $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n) \in (\mathbb{T}^d)^n$. The corresponding target measure is

$$\bar{\mu}^{(n)} := \sum_{i=1}^n s_i \delta_{\bar{x}_i}.$$

A convenient weighted Gaussian terminal condition on the universal cover $(\mathbb{R}^d)^n$ is

$$(3.32) \quad g_{n,\lambda}(s, x) := \exp\left(-\frac{\lambda}{2} \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2\right), \quad \lambda > 0.$$

This is the natural Gaussian centred at the configuration $(s, \bar{x})$ for the weighted geometry induced by the generator $L_s^{(n)} = \sum_i s_i^{-1} \Delta_{x_i}$. On the torus one may either periodize (3.32), or work on a fixed lift as long as wrap-around effects are negligible on the time interval of interest.

We now derive the corresponding closed form for $u_t^{(n)}$ in detail. Set

$$\tau := T - t, \quad v_\tau^{(n)}(s, x) := u_{T-\tau}^{(n)}(s, x).$$

Since $u_t^{(n)}$ solves the backward equation

$$\partial_t u_t^{(n)} + \sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} u_t^{(n)} = 0, \quad u_T^{(n)} = g_{n,\lambda},$$

the function $v_\tau^{(n)}$ solves the forward heat equation

$$(3.33) \quad \partial_\tau v_\tau^{(n)} = \sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_\tau^{(n)}, \quad v_0^{(n)} = g_{n,\lambda}.$$

Because both the generator and the terminal datum factorize over the particle labels, one expects the solution to keep the same Gaussian shape. Writing

$$R(x) := \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2,$$

we therefore make the ansatz

$$(3.34) \quad v_\tau^{(n)}(s, x) = A(\tau) \exp\left(-\frac{B(\tau)}{2} R(x)\right),$$

with initial conditions $A(0) = 1$ and $B(0) = \lambda$ so that $v_0^{(n)} = g_{n,\lambda}$.

Differentiating (3.34) with respect to τ gives

$$\partial_\tau v_\tau^{(n)} = \left(\frac{A'(\tau)}{A(\tau)} - \frac{B'(\tau)}{2} R(x) \right) v_\tau^{(n)}.$$

For each $i = 1, \dots, n$ we also have

$$\nabla_{x_i} v_\tau^{(n)} = -B(\tau) s_i (x_i - \bar{x}_i) v_\tau^{(n)},$$

and therefore

$$\Delta_{x_i} v_\tau^{(n)} = \left(-B(\tau) s_i d + B(\tau)^2 s_i^2 |x_i - \bar{x}_i|^2 \right) v_\tau^{(n)}.$$

Summing with the weights $1/s_i$ from the generator gives

$$\sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_\tau^{(n)} = \left(-nd B(\tau) + B(\tau)^2 \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2 \right) v_\tau^{(n)} = (-nd B(\tau) + B(\tau)^2 R(x)) v_\tau^{(n)}.$$

Substituting this identity and the expression for $\partial_\tau v_\tau^{(n)}$ into (3.33) and identifying the coefficients of 1 and $R(x)$ yields the ODE system

$$(3.35) \quad \frac{A'(\tau)}{A(\tau)} = -nd B(\tau), \quad -\frac{1}{2}B'(\tau) = B(\tau)^2, \quad A(0) = 1, \quad B(0) = \lambda.$$

The second ODE is scalar and can be solved explicitly:

$$B'(\tau) = -2B(\tau)^2 \quad \implies \quad B(\tau) = \frac{\lambda}{1 + 2\lambda\tau}.$$

Plugging this into the first ODE gives

$$\log A(\tau) = -nd \int_0^\tau \frac{\lambda}{1 + 2\lambda r} dr = -\frac{nd}{2} \log(1 + 2\lambda\tau),$$

so that

$$A(\tau) = (1 + 2\lambda\tau)^{-nd/2}.$$

Substituting A and B back into (3.34) and recalling that $\tau = T - t$ yields the closed formula

$$(3.36) \quad u_t^{(n)}(s, x) = (1 + 2\lambda(T - t))^{-nd/2} \exp\left(-\frac{\lambda}{2(1 + 2\lambda(T - t))} \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2\right).$$

As a consistency check, when $t = T$ the prefactor equals 1 and (3.36) reduces exactly to the terminal condition (3.32).

Taking logarithms in (3.36),

$$\log u_t^{(n)}(s, x) = -\frac{nd}{2} \log(1 + 2\lambda(T - t)) - \frac{\lambda}{2(1 + 2\lambda(T - t))} \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2.$$

Hence, for each i ,

$$\nabla_{x_i} \log u_t^{(n)}(s, x) = -\frac{\lambda s_i}{1 + 2\lambda(T - t)} (x_i - \bar{x}_i),$$

and therefore

$$(3.37) \quad D_i^{(n)} \log u_t^{(n)}(s, x) = \frac{1}{s_i} \nabla_{x_i} \log u_t^{(n)}(s, x) = -\frac{\lambda}{1 + 2\lambda(T - t)} (x_i - \bar{x}_i), \quad b_i^{(n)}(t, x; s) = -\frac{2\lambda}{1 + 2\lambda(T - t)} (x_i - \bar{x}_i).$$

Thus each particle satisfies the time-inhomogeneous Ornstein–Uhlenbeck-type SDE

$$(3.38) \quad d\xi_i^{(n)}(t) = -\frac{2\lambda}{1 + 2\lambda(T - t)} (\xi_i^{(n)}(t) - \bar{x}_i) dt + \sqrt{\frac{2}{s_i}} d\widehat{\beta}_i^{(n)}(t).$$

For this weighted Gaussian choice, the restoring drift is independent of s_i , while the noise intensity still depends on $s_i^{-1/2}$.

Algorithm 1 Euler–Maruyama scheme for the Gaussian terminal condition centred at $\bar{\mu}^{(n)}$

Require: n , masses $s = (s_1, \dots, s_n)$, target locations $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n)$, parameter $\lambda > 0$, horizon T , step size Δt , and initial configuration $X^0 \in (\mathbb{T}^d)^n$

Ensure: Approximate path $(X^m)_{m=0}^M$ and measures $\mu_{t_m}^{(n)} = \sum_{i=1}^n s_i \delta_{X_i^m}$

```

1:  $M \leftarrow T/\Delta t$  and  $t_m \leftarrow m\Delta t$  for  $m = 0, \dots, M$ 
2: for  $m = 0, \dots, M - 1$  do
3:    $\tau_m \leftarrow T - t_m$ 
4:   for  $i = 1, \dots, n$  do
5:     Either use the Euclidean closed form
           
$$b_i^m \leftarrow -\frac{2\lambda}{1 + 2\lambda\tau_m}(X_i^m - \bar{x}_i),$$

           or use the wrapped drift obtained from a truncated torus image sum
6:     Draw  $Z_i^m \sim N(0, I_d)$  independently
7:      $X_i^{m+1} \leftarrow X_i^m + b_i^m \Delta t + \sqrt{\frac{2\Delta t}{s_i}} Z_i^m$ 
8:     Reduce  $X_i^{m+1}$  modulo  $\mathbb{Z}^d$ 
9:   end for
10: end for
11: end for

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